D3.1 Reproduction (Plant)

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/sl-d3-1-plant.html

Written for this site from the course textbook and class slides. Use the page numbers to read the book's fuller account. Human reproduction (D3.1.1–7) is in Body Systems. Image credits are on the last slide of the slides for this subtopic.

FROM FLOWER TO SEED

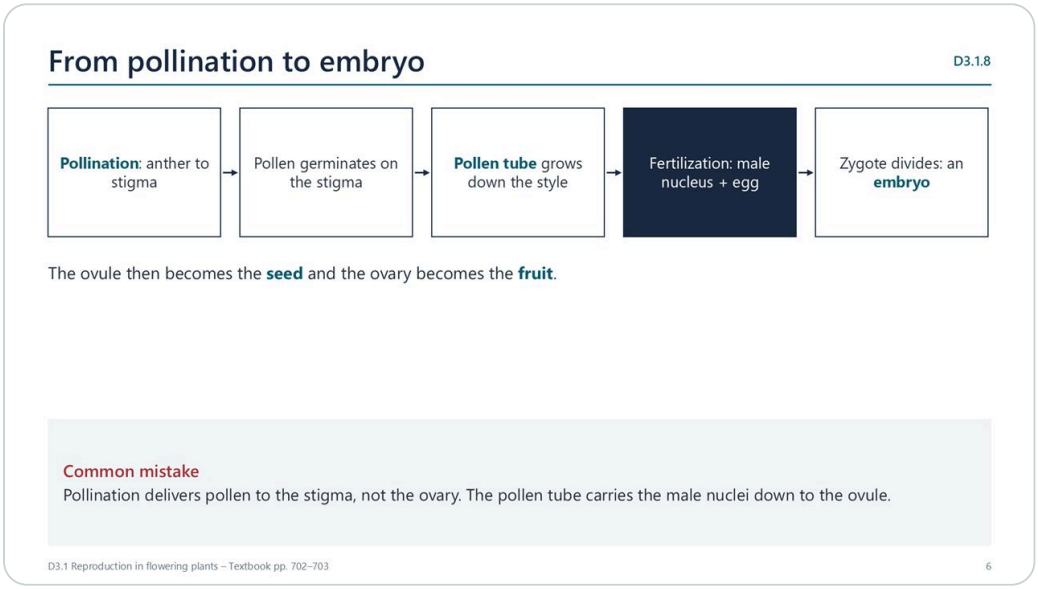
D3.1.8 How flowering plants reproduce sexually

pp. 702–703

Male gametes develop inside **pollen grains**, and the egg develops inside an **ovule** in the ovary.

Many flowers are hermaphrodite, yet reproduction is still sexual, because gametes fuse.

1. **Pollination:** pollen moves from an anther to a stigma.
2. The pollen germinates, and a **pollen tube** grows down the style to the ovule.
3. **Fertilization:** a male nucleus fuses with the egg, forming a zygote. The other male nucleus triggers the formation of a food store.
4. The zygote divides by mitosis to form an **embryo**. The ovule becomes the seed, and the ovary the fruit.



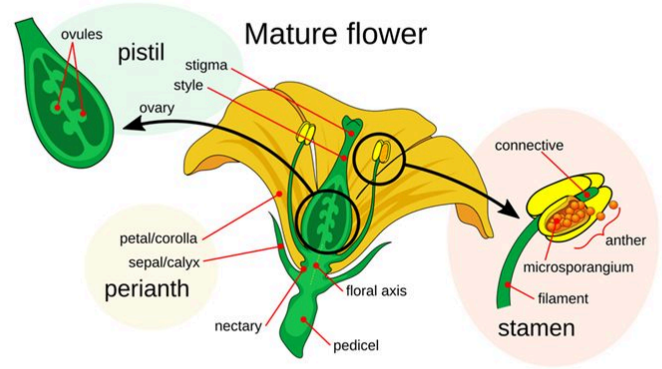
Watch out: pollination delivers pollen to the stigma, not to the ovary.

D3.1.9 The features of a flower pollinated by insects

p. 703

An insect-pollinated flower

D3.1.9



A mature flower. Mariana Ruiz, LadyofHats, public domain.

Part	Function
Sepal	Protects the bud
Petal	Attracts pollinators
Anther	Makes pollen
Filament	Holds up the anther
Stigma	Pollen lands here
Style	Holds up the stigma
Ovary	Contains ovules
Nectary	Nectar as a reward

Exam note
Stamen = anther + filament. Carpel = stigma + style + ovary. Be ready to draw and annotate.

Sepals protect the bud. Colourful **petals** and sugary **nectar** attract pollinators. The **stamen** is an anther, which makes pollen, on a filament. The **carpel** is a stigma, where pollen lands, with a style and an ovary containing ovules.

AVOIDING SELF-POLLINATION

D3.1.10 Ways of encouraging cross-pollination

pp. 704–705

Cross-pollination moves pollen between plants of the same species, often by animals or wind. It increases variation. Plants encourage it in these ways:

- Pollen and stigma mature at different times.
- Separate male and female flowers grow on one plant, as in birch.
- Separate male and female plants exist, as in holly.

D3.1.11 How self-incompatibility raises genetic variation

p. 705

Self-pollination leads to **inbreeding**, which decreases genetic diversity and vigour. In many species, genes let a plant recognise its own pollen. If the pollen carries the same self-incompatibility allele as the style, the pollen tube stops growing. So gametes that fuse come from different plants.

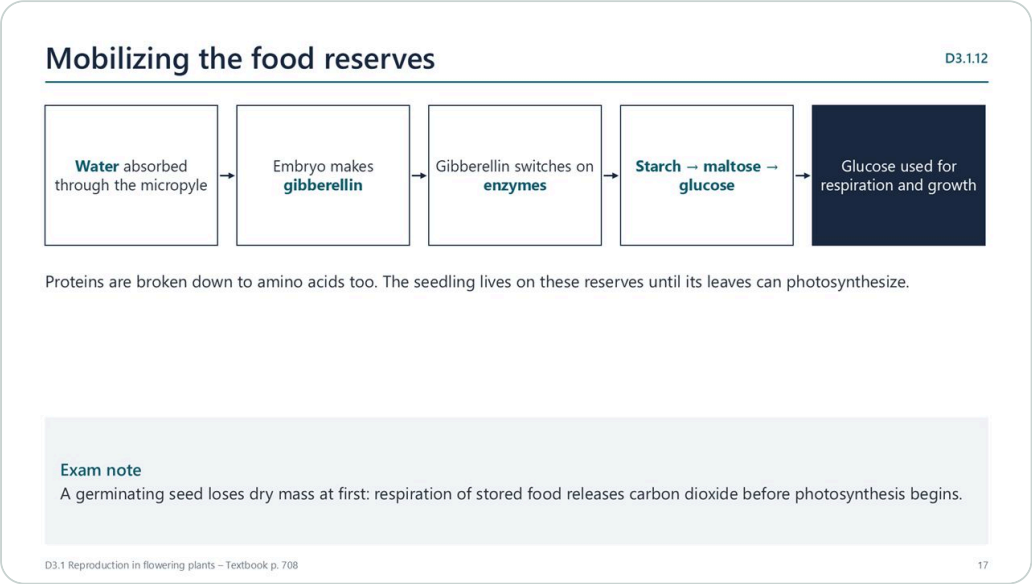
D3.1.12 How seeds disperse and germinate

pp. 706–708

Dispersal moves seeds away from the parent by wind, water, animals or explosive pods. This reduces competition for space, light, nutrients and water.

Germination needs water to rehydrate the cells, oxygen for aerobic respiration, and a temperature suitable for enzymes.

1. Water enters through the micropyle.
2. The embryo produces **gibberellin**.
3. Gibberellin stimulates enzymes in the food store.
4. Starch is hydrolysed to maltose, and then glucose.
5. Glucose and amino acids move to the growing points, for respiration and growth.



Watch out: pollination and dispersal are different events. Pollination comes before fertilization, and dispersal after it.

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